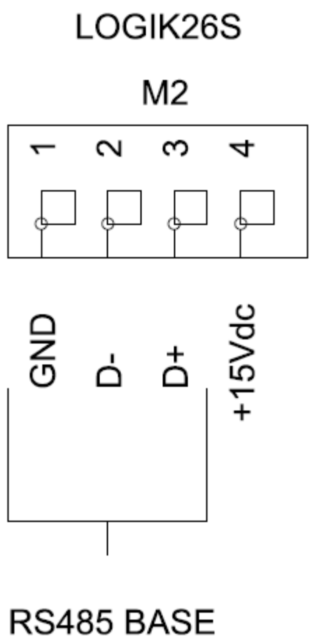
CONTENTS

SERIAL CONNECTIONS

The serial communication type is RS485 and uses modbus RTU protocol.



Each logik26S has 2 serial communcation ports on the termials M2(Master/slave - Multinuit) and M8(Inverter or stand alone comm. only)

Master / Slave operation only two compressor must be connected via serial communication.In case of communication between two units. no need to use 120 Ω resistor for each end.

Serial Connection Notes

1. Use shielded flexible twisted cable
2. Connect D+ and D- terminals through each compressor controller and connect shield part to terminal "M2-1 : GND" digital ground for each side as well.
3. Do not connect the cables shield to the ground
4. The cable lenght must not be longer than 400 meters
5. The units on the serial net must be connected without any reversal connection and connections must be like :

   D+ -> D+ / D- -> D- / 0 -> 0

Note :

The serial port on the terminal M2 can be used for ;

- **Master / Slave Operation usage**
- **Stand alone modbus communications (Scada Systems, modbus softwares on pc etc.)**
- **Multi unit Operation**

The serial port on the terminal M8 can be used for inverter communications or for stand alone communication.

MODBUS COMMUNICATION

The serial connection has described in Serial Connection Section.

Logik26S has 1 RS485 Serial interface for modbus communication with external systems. This serial interface supports modbus RTU protocol and allow to read compressor data's under the conditions :

- Compressors must be working stand alone. If two compressors are working under master / slave mode, its not possible to use same serial line for external communication.
- If compressors are working under the multiunit operation, then Logika ethernet serial converter interface is needed to make possible to read data from the compressors externally via modbus tcp protocol. (Serial ethernet interface also have capability to convert modbus rtu into the modbus tcp.)
  Note: Its not possible to use standard serial ethernet converter to convert the modbus rtu protocol into the modbus tcp, since the logik ethernet serial interface has special function to prevent master message interruption. Standard converters do not have such capability.

Modbus communication parameters :

Baud Rate : 9600

Data Bits : 8

Stop Bits : 1

Parity : None

These modbus connection parameters are constant and they are not able to change.

Each compressor number is also modbus address of the relevant compressor. All the compressors must have different compressor number under the compressor number parameter setting C08.

MODBUS ADDRESS LIST FOR THE LOGIK 26S

Group 0:System Identification					
Word	**Type**	**Read/Write**	**Len**	**Description**	**Notes**
0000	char[20]	RW	20	Serial Number	ASCII string '\0' terminated
000A	word	R	2	Logika Control MODBUS controller model	Logik26S model coded as follow: 0x02A1 Dalgakiran version
000B	char[2]	R	2	Firmware Release	Low byte= minor release, High byte = major release
000C	word	R	2	MODBUS Interface Release	This table applies to 0x0001

Group 1: Passwords					
Word	**Type**	**Perm.**	**Size**	**Description**	**Notes**
0100	char[6]	RW	6	Level 1 Password	The first 2 characters are significant. Admitted values are ASCII from '0' to '9'
0103	char[6]	RW	6	Level 2 Password	The first 4 characters are significant. Admitted values are ASCII from '0' to '9'
0106	char[6]	RW	6	Level 3 Password	All the characters are significant. Admitted values are ASCII from '0' to '9'

Group 2: Alarm active and acknowledged					
Word	**Type**	**Read/Write**	**Len**	**Description**	**Notes**
0200	byte[8]	R	8	Active Alarms	Bit mapped allocation. Starting from bit 1 of the first byte (bit 0 is not used), each bit indicates that the corresponding alarm in the table "Alarm List" is active
0204	byte[8]	R	8	Not User Acknoledged Alarms	Bit mapped allocation with the same allocation as "Active Alarms". When an alarm is raise the correspondent bit is set in this structure. This bit will reset when the user press the RESET key (if the alarm requires user reset) or when the alarm cause will end (alarms with automatic reset).

Group 3: Alarm records					
Word	**Type**	**Read/Write**	**Len**	**Description**	**Notes**
0300	byte[2]	RW	2	Indexes	The structure is used in circular manner, if an alarm fills the last position available, 19, the new alarm will be recorded in position 0. - Byte L: Position Index for the new alarm that will be recorded (0..19) - Byte H: Number of significant alarms (0..20)
0301	byte[100]	R	100	20 Alarm Records	A single record is made of 5 bytes: - 4 bytes for the system time, from the least significant bit: 6 bit code minute 5 bit code hour 5 bit code day of the month 4 bit code month 7 bit code year (from 0 to 99, 0 is 2000) - 1 byte alarm code (see table "Alarm List")

Group 4: Controller State and Measures					
Word	**Type**	**Read/Write**	**Len**	**Description**	**Notes**
0400	word	R	2	Internal Controller State	By values: 0 : RESET 1 : OFF 2 : STARTING, MOTOR IN STAR CONNECTION 3 : STARTING, PAUSE STAR TO DELTA CONNECTION 4 : STARTING, ACCELERATING IN DELTA CONNECTION 5 : LOAD RUNNING 6 : IDLE RUNNING, PRESSURE IN RANGE 7 : IDLE RUNNING, STOPPING 8 : INVERTER ON 9 : INVERTER SETUP 10 : BLOCKED BY FAULT 11 : FACTORY TEST
0401	word	R	2	Displayed state	By values: 0 : OFF 1 : INTERNAL PRESS TOO HIGH, WAITING 2 : REMOTE STOP ACTIVE 3 : STOP BY TIMER 4 : IDLE STOPPING 5 : IDLE STOPPING BY REMOTE STOP 6 : IDLE STOPPING BY TIMER 7 : PRESSURE IN SET, MOTOR IS OFF 8 : WAITING TO START (security timer Wt5) 9 : MOTOR STARTING 10 : IDLE RUNNING 11 : LOAD RUNNING 12 : SOFT BLOCK DELAY (30 SECONDS) 13 : BLOCK 14 : FACTORY TEST
0402	word	R	2	Code of alarm currently blocking or 0x00	Codes are reported on Logik26S manual or in the table "Alarm List"
0403	word	R	2	Rele Output	Bit mapped allocation: 0x0001 RL1 0x0002 RL1 0x0004 RL3 0x0008 RL4 0x0010 RL5 0x0020 RL6 0x0040 RL7

0404	word	R	2	Digital Input	Bit mapped allocation: 0x0001 IN1 0x0002 IN2 0x0004 IN3 0x0008 IN4 0x0010 IN5 0x0020 IN6 0x0040 IN7 0x0080 Phase R (Phase Check Unit Logika Control) 0x0100 Phase S (Phase Check Unit Logika Control) 0x0200 Phase T (Phase Check Unit Logika Control) 0x0400 INVERTER FAULT INPUT 0x0800 PTC state 0x1000 Phases Sequence Correct (Phase Check Unit Logika Control)
0405	int	R	2	Screw Temperature	In Celsius * 10
0406	word	R	2	Working Pressure	In bar * 10
0407	word	R	2	Auxiliary Pressure	In bar * 10
0408	word	R	2	PTC Input	In Volt * 1000
0409	word	R	2	Controller Supply Voltage (nominal +15V)	In Volt * 10
040A	word	R	2	Controller status indicators	Bit mapped allocation: 0x0001 COMPRESSOR ACTIVE (ON) 0x0002 COMPRESSOR MASTER (WHEN MASTER/SLAVE ACTIVE) 0x0004 COMPRESSOR ON BY WEEKLY TIMER 0x0008 WEEKLY TYMER BYPASSED BY USER 0x0010 - 0x0020 FAN LOGICAL OUT 0x0040 DRAIN LOGICAL OUT 0x0080 ALARM LOGICAL OUT 0x0100 REMOTE START/STOP (0 STOP) 0x0200 MOTOR IS STARTING 0x0400 COMPRESSOR IS ACTIVE 0x0800 MOTOR RUNNING 0x1000 ACTING AS MULTIUNIT SLAVE 0x2000 COMPRESSOR CONTROLLED BY MULTIUNIT MASTER 0x4000 MULTIUNIT ON/OFF BY WEEKLY TIMER 0x8000 MULTIUNIT MASTER FAULT

040B	word	R	2	Analog out frequency set	Hz
040C	word	W	2	Controller fieldbus commands	Bit mapped allocation: 0x0001 START COMPRESSOR 0x0002 STOP COMPRESSOR 0x0004 ALARM RESET 0x0008 START COMPRESSOR BYPASSING WEEKLY TIMER 0x0010 STOP COMPRESSOR BYPASSING WEEKLY TIMER 0x0020 ACK & RESET ALL ALARMS 0x0040 LOAD COMPRESSOR, VALID ONLY IF BIT B15(0x8000) IS SET 0x0080 WATCHDOG (IF SET, MUST BE SET AGAIN EVERY 5 SECONDS OTHERWISE COMPRESSOR STOPS WITH A FIELDBUS FAULT THAT MUST BE RESET EITHER BY FIELDBUS OR MANUALLY) 0x0100 RESET AIR FILTER MAINT. COUNTER 0x0200 RESET OIL FILTER MAINT. COUNTER 0x0400 RESET SEPARATOR FILTER MAINT. COUNTER 0x0800 RESET OIL MAINT. COUNTER 0x1000 RESET COMPRESSOR MAINT. COUNTER 0x2000 RESET BEARING MAINT. COUNTER 0x4000 TO BE DEFINED, VALID ONLY IF BIT B15(0x8000) IS SET 0x8000 THIS BIT ENABLES BIT B6(0x4000) AND BIT B14(0x4000). IF THIS BIT IS SET, THE COMMANDS FROM B6 AND B7 ARE VALID FOR THE NEXT 10 SECONDS.
040D	word	W	2	Relative speed (for delivery calculation for VSD compressor)	0..1000 (maximum speed)

Group 5: Parameters					
Word	**Type**	**Read/Write**	**Len**	**Description**	**Notes**
0500	word	RW	2	Configuration switches	Bit mapped allocation (see menu Compressor Config): 0x0001 C01 – Automatic Restart 0x0002 C03 – Fixed Timer Wt4 0x0004 C04 – Phase Check Active 0x0008 C05 – Security Active 0x0010 C06 – Low supply voltage check active 0x0020 C11 – PTC input enabled 0x0040 Temperature in Fahrenheit (menu M1-3) 0x0080 Pressure in PSI (menu M1-3) 0x0100 DST automatic adjust (menu M1-3) 0x0200 T01 - Start/stop by weekly timer enabled 0x0400 C17 – Block compressor on expiring of C–H maintenance 0x0800 C07.3 – Maintenance mode when working under multiunit 0x1000 C07.4 – Both inverter varying speed when master/slave new is active

0501	int	RW	2	Configuration selections 1 / 2	Bit mapped allocation (see menu Compressor Config): Bit0..Bit3: C13 - RL2 mode: Bit4..Bit7: C14 - RL5 mode: Bit8..Bit11: C15 - RL6 mode Bit12..Bit15: C16 - RL7 mode 0 default, 1 fan, 2 drain, 3 state, 4 alarm, 5 motor, 6 load, 7 lubricate, 8 null, 9 PORO, 10 PORO + Alarm
0502	int	RW	2	Configuration selections 2 / 2	Bit mapped allocation (see menu Compressor Config): Bit0..Bit2: C19 – AUX PRESS INPUT 0 disabled, 1 delta pressure, 2 power from drive, 3 current from drive, 4 temperature from drive Bit3..Bit4: C07 – MULTIUNIT 0 no multiunit or master/slave, 1 master/slave 2 master/slave new, 3 multiunit slave Bit5..Bit6: C18 – ANALOG OUTPUT MODE 0 not used, 1 regulate on pressure, 2 regulate on temperature Bit7..Bit9: C12 – IN7 CONFIGURATION 0 disabled, 1 door, 2 control phase relay, 3 generic alarm, 4 bearing high temp alarm Bit10..Bit11: C21 – INVERTER FAULT INPUT 0 disabled, 1 normally open, 2 normally closed
0503	int	RW	2	LANGUAGE	Values: 0 ITALIAN 1 ENGLISH
0504	word	RW	2	Level 1 Password	Password level 1 BCD coded (only first 2 digits significant)
0505	word	RW	2	Level 2 Password	Password level 2 BCD coded (4 digits)
0506	word	RW	2	Level 3 Password 1 / 2	Password level 3 BCD coded (first 4 digits)
0507	word	RW	2	Level 3 Password 2 / 2	Password level 3 BCD coded (last 2 digits)
0508	word	RW	2	LCD contrast	Default 250, Set between 200 and 380
0509	int	RW	2	WP1	bar
050A	int	RW	2	WP2	bar*10
050B	int	RW	2	WP3	bar*10
050C	int	RW	2	WP4	bar*10
050D	int	RW	2	WP5	bar*10
050E	int	RW	2	WP6	bar*10
050F	int	RW	2	WT0	0 disable/1 KTY/2 NTC

0510	int	RW	2	WT1	°C
0511	int	RW	2	WT2	°C
0512	int	RW	2	WT3	°C
0513	int	RW	2	WT4	°C
0514	int	RW	2	WT5	°C
0515	int	RW	2	WT6	°C
0516	int	RW	2	WT7	°C
0517	int	RW	2	Wt1	sec
0518	int	RW	2	Wt2	msec
0519	int	RW	2	Wt3	sec
051A	int	RW	2	Wt4	min
051B	int	RW	2	Wt5	sec
051C	int	RW	2	Wt6	sec
051D	int	RW	2	Wt7	min
051E	int	RW	2	C07.1	hour
051F	int	RW	2	C07.2	min
0520	int	RW	2	CAF SET	hour
0521	int	RW	2	COF SET	hour
0522	int	RW	2	CSF SET	hour
0523	int	RW	2	C– SET	hour
0524	int	RW	2	C–h SET	hour
0525	int	RW	2	C–BL SET	hour
0526	int	RW	2	C08	MODBUS ADDRESS
0527	int	RW	2	C02	MAX STARTS PER HOUR
0528	int	RW	2	C10	L/min*0.1
0529	int	RW	2	AP1	bar*10
052A	int	RW	2	AP2	bar*10
052B	int	RW	2	AP3	bar*10
052C	int	RW	2	C19.1	sec
052D	int	RW	2	AP4	bar*10
052E	int	RW	2	C19.2	Range analog input
052F	int	RW	2	PI1	0,01
0530	int	RW	2	PI2	0,01
0531	int	RW	2	PI3	0,01

0532	int	RW	2	PI4	0,01
0533	int	RW	2	PI5	0,01
0534	int	RW	2	PI6	0,01
0535	int	RW	2	PI7	0,01
0536	int	RW	2	FR1	Hz
0537	int	RW	2	FR2	Hz
0538	int	RW	2	PT1	sec*10
0539	int	RW	2	PT2	sec*10
053A	int	RW	2	PT3	sec*10
053B	int	RW	2	PM1	0,01
053C	int	RW	2	AO1	bar*10
053D	int	RW	2	AO2	bar*10
053E	int	RW	2	AO3	A*10
053F	int	RW	2	AO4	A*10
0540	int	RW	2	C20.1	sec
0541	int	RW	2	C20.2	sec
0542	int	RW	2	C22	sec
0543	int	RW	2	C23	min
0544	int	RW	2	DR0	Drive protocol: 0 NO INVERTER CONNECTED ON SECOND RS485 1 DANFOSS FC
0545	int	RW	2	DR1	[Hz]
0546	int	RW	2	DR2	[Hz]
0547	int	RW	2	DR3	[0.1s]
0548	int	RW	2	DR4	[0.1s]
0549	int	RW	2	DR5	[0.01]
054A	int	RW	2	DR6	[0.01]
054B	int	RW	2	DA0	Depends on protocol, see specific application document for drive selected
054C	int	RW	2	DA1	Depends on protocol, see specific application document for drive selected
054D	int	RW	2	DA2	Depends on protocol, see specific application document for drive selected
054E	int	RW	2	DA3	Depends on protocol, see specific application document for drive selected
054F	int	RW	2	DA4	Depends on protocol, see specific application document for drive selected
0550	int	RW	2	DA5	Depends on protocol, see specific application document for drive selected
0551	int	RW	2	DA6	Depends on protocol, see specific application document for drive selected
0552	int	RW	2	DA7	Depends on protocol, see specific application document for drive selected
0553	int	RW	2	DA8	Depends on protocol, see specific application document for drive selected
0554	int	RW	2	DA9	Depends on protocol, see specific application document for drive selected

Group 6: Counters					
Word	**Type**	**Read/Write**	**Len**	**Description**	**Notes**
0600	long	RW	4	Number total compressor hour (Rele' RL1 active)	In minutes
0602	long	RW	4	Number on load compressor hour (Rele' RL4 active)	In minutes
0604	long[6]	RW	24	Maintenaince Counter Count (time from last mantenaince), by index: 0 - AIR FILTER 1 - OIL FILTER 2 - SEPARATOR FILTER 3 - OIL 4 - COMPRESSOR 5 – BEARING LUBRICATE	In minutes
0610	word	R	2	Percentage or on load hours vs total	On load minutes in last 100hours of motor running: 100% is 6000
0611	word	R	2	Number of starts in the last hour	

Group 7: Clock timer start/stop					
Word	**Type**	**Read/Write**	**Len**	**Description**	**Notes**
0700	byte[84]	RW	84	Clock Timers	7 (number of day in a week) x 3 (number of clock timer per day) structure of 4 bytes giving: -minutes start -hour start -minute stop -hours stop (if last timer of a day falls in the next day the stop hour is incremented by 24)

Group 8: Time set					
Word	**Type**	**Read/Write**	**Len**	**Description**	**Notes**
0800	byte[8]	RW	8	System time	7 bytes array, by index : 0 - SECONDS 1 - MINUTES 2 - HORS, 3 - DAY OF WEEK (1=MONDAY) 4 - DAY OF MONTH 5 - MONTH 6 - YEAR (00-99, 00=2000) Last byte is not used
0804	word	R	2	Timekeeper error flags	Bit mapped allocation: 0x0001 Read error 0x0002 Write error 0x0004 Signature error 0x0008 No clock oscillation 0x0010 Read/Write error 0x0020 Generic fault
0805	word	R	2	Daylight Saving Time adjustment	Two values: -1 legal time -0 solar time

Group 9: Maintenaince records					
Word	**Type**	**Read/Write**	**Len**	**Description**	**Notes**
0900	byte[2]	RW	2	Indexes	The structure is used in circular manner as alarm records. - Byte L: Position Index for the new maintenance event that will be recorded (0..19) - Byte H: Number of significant events (0..20)
0901	byte[120]	RW	120	20 Maintenance Operation Records	A single record is made of 6 bytes: - 4 bytes for the system time, from the least significant bit: 6 bit code minute 5 bit code hour 5 bit code day of the month 4 bit code month 7 bit code year (from 0 to 99, 0 is 2000) 3 bit code the counter resetted: 0 CAF, 1 COF, 2 CSF, 3 C__, 4 C__h, 5 C-BL - 2 bytes (integer value) count down value when reset occured (in hours)

CODE	ALARM
1	A01-EMERGENCY
2	A02-MOTOR OVERHEAT
3	A03-FAN OVERHEAT
4	A04-AC PHASE MISSING
5	A05-AC PHASE SEQ WRONG
6	
7	A07-DOOR OPEN
8	
9	A09-DRIVE FAULT
10	
11	A11-HIGH WORK PRESS
12	A12-SCREW TEMP FAULT
13	A13-HIGH SCREW TEMP
14	A14-LOW SCREW TEMP
15	A15-AUX TRANSD SEP FILTER
16	
17	
18	A18-BLACK OUT
19	
20	A20-PTC MOTOR
21	A21-INPUT COMMON MISSING
22	A22-INPUT7
23	
24	
25	A25-SEPARATOR FILTER
26	A26-WORK PRESS FAULT
27	A27-AUX PRESS FAULT
28	A28-LOW VOLTAGE
29	A29-SECURITY
30	A30-SCREW TEMP WARN
31	

CODE	ALARM
32	A32-MAINT C H BLK
33	A33-FIELDBUS ERR
34	
35	A35-EEPROM FAULT
36	A36-AIR FILTER
37	A37-MU FAULT
38	A38-AUX TRANSD SEP FILTER WARN
39	A39-LOW VOLTAGE WARN
40	A40-HIGH VOLTAGE
41	A41-TIMEKEEPER FAULT
42	A42-RS232 FAULT
43	A43-DST ADJUSTED
44	A44-BEARING HIGH TEMP
45	
46	
47	A47-TOO MUCH START
48	A48-RESTART MANUAL
49	A49-RESTART AUTO
50	A50-MAINT CAF
51	A51-MAINT COF
52	A52-MAINT CSF
53	A53-MAINT C--
54	A54-MAINT C-H
55	A55-MAINT BL
56	
57	
58	
59	
60	A60-DRIVE FAULT
61	A61-DRIVE WARNING
62	A62-DRIVE NO COMMUNICATION